# Mahmoud Alyosify

## Computer Vision Engineer

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## SUMMARY

Computer vision engineer focused on image models that **learn with few labels and are ready to deploy**. Built a self-supervised image model (SimCLR) that reaches 84.30% top-1 accuracy and serves image search through ONNX and FAISS, and led a system that reads vital signs from ordinary video (Microsoft Imagine Cup 2023 semi-finalist). MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), with graduate coursework and certification in deep learning.

## SKILLS

**Computer vision:** CNNs, image classification, self-supervised and contrastive learning (SimCLR, SupCon), image retrieval, data augmentation
**Video & signals:** rPPG, video signal processing, spectrograms
**Tools:** PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Matplotlib
**Deployment:** ONNX, FAISS, FastAPI, Streamlit, Docker, AWS
**Programming:** Python, C++, SQL, Git, Linux

## PROJECTS

### SimCLR-Vision-SSL — self-supervised image model
2026
*PyTorch, SimCLR, SupCon, ONNX, FAISS*

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

### Vitalism Solution — graduation project
2023
*Python, computer vision, signal processing (rPPG)*

- Measures heart rate and other vital signs from ordinary video, with no contact sensors. Grade **A+**; **Microsoft Imagine Cup 2023 semi-finalist**.
- Led a six-student team from idea to a working prototype; Finalist at ISEIC 2023.

### AudioShield — deepfake audio detection
*Python, CNN, Streamlit*

- Trained a CNN on spectrograms to tell real speech from synthetic speech; served it in a Streamlit app with live predictions and PDF reports.

### SCRAP — satellite collision risk prediction
2026
*Python, XGBoost, LightGBM, Optuna, SHAP*

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.

Also: MinimalLM (MSc thesis, 47.6% fewer LLM tokens) · Horus-OSINT (Llama-3-8B fine-tuned on AWS) · HORUS Sentinel (multi-agent OSINT).

## EXPERIENCE

**AI Engineer Intern**
Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy)
Maadi, Cairo

- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

**Part-Time Machine Learning Instructor**
Jul 2025 – Present
National Telecommunication Institute (NTI)
Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.

**AI in Cybersecurity Intern** Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

**Systems Engineer (Military Service)** Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.

**Online Instructor** 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

## PUBLICATIONS

**Say Less, Mean Better: Training-Free Verbosity Control for Language Models**
*M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026*

**Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding**
*M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026*

## EDUCATION

**MSc in Artificial Intelligence** Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

**BSc in Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

## CERTIFICATIONS

- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- IBM AI Developer Professional Certificate (IBM, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- HCIP-AI training course (Huawei, 80 hours, 2025)
- Machine Learning Internship, ITIDA / Egypt Makes Electronics (200 hours, 2023)

## AWARDS & LEADERSHIP

- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Ideal Student Award, Faculty of Computers & Information, Assiut University (2022–2023)

## LANGUAGES

Arabic (native) · English (professional working proficiency)